

HWK 1
APPLIED NATURAL LANGUAGE PROCESSING
DATA 641

Converting Jupyter Notebooks to HTML

To convert your Jupyter Notebook to an HTML file, follow these steps:

1. Open the Jupyter Notebook that you want to convert to an HTML file.
2. Ensure that the notebook is saved with the latest changes.
3. In the Jupyter Notebook interface, navigate to **File** in the menu bar.
4. Click on **Download as** and select **HTML (.html)** from the dropdown menu. This will initiate the conversion process.
5. The Jupyter Notebook will be converted to an HTML file and automatically downloaded to the default download location on your computer.
6. Locate the downloaded HTML file on your computer and verify that the conversion was successful.
7. Optionally, open the HTML file in a web browser to view and verify the content.
8. Upload your HTML file on Canvas.

Make sure to save your Jupyter Notebook before attempting to convert it to an HTML file.

Exercise 1 (6 points)

Programming Fundamentals and Python (https://courses.ischool.berkeley.edu/i256/f06/papers/programming_tutorial.pdf). From this file answer the following questions (exercises from the web reading):

- 2.3.4 – 1, 2 (0.5 pt. each – 1 points total)
- 2.4.5 – 1, 2, 3, 4 (0.5 pts. each – 2 pts. total)
- 2.5.4 – 1, 2 (1 pts. each – 2 pts. total)
- 2.6.4 – 1 (1 pt.)

Answer each individual question (e.g., 2.3.4 question 2 b) in one notebook cell.

Exercise 2 (3 points)

Implement the following function:

```
#function squaring_dictionary
#     input: n, an integer
#     output: sq_dict, a dictionary with keys from (1...n),
#             with values equal to n**2
#
#     Example: squaring_dictionary(4)
#             returns {1:1, 2:4, 3:9, 4:16}

def squaring_dictionary(n):
# ... place implementation here

return sq_dict
```

Exercise 3 (6 points)

Copy the following code to your notebook, and implement the below functionalities.

```
lloyd = {
    "name": "Lloyd",
    "homework": [90.0, 97.0, 75.0, 92.0],
    "quizzes": [88.0, 40.0, 94.0],
    "tests": [75.0, 90.0]
}
alice = {
    "name": "Alice",
    "homework": [100.0, 92.0, 98.0, 100.0],
    "quizzes": [82.0, 83.0, 91.0],
    "tests": [89.0, 97.0]
}
tyler = {
    "name": "Tyler",
    "homework": [0.0, 87.0, 75.0, 22.0],
    "quizzes": [0.0, 75.0, 78.0],
    "tests": [100.0, 100.0]
}
```

- (a) Write a function that returns averages of the homework, quizzes and tests of dictionaries with the above keys.

```
#function work_averages
#     input: in_dict, a dictionary with keys:
```

```

#             homework: list of floats
#             quizzes: list of floats
#             tests: list of floats
# output: None. Prints out averages of homework,
#           quiz, and test scores
#
# Example: work_averages(tyler)
#             >> Tyler averages:
#             >> homework: 46.0
#             >> quizzes: 51.0
#             >> tests: 100.0

def work_averages(in_dict):
# ... place implementation here

```

- (b) Write a function that returns the lowest homework, quiz and test score of dictionaries with the above keys.

```

#function lowest_score
# input: in_dict, a dictionary with keys:
#             homework: list of floats
#             quizzes: list of floats
#             tests: list of floats
# output: None. Prints out lowest homework,
#           quiz, and test scores
#
# Example: lowest_score(tyler)
#             >> Tyler averages:
#             >> homework: 0.0
#             >> quizzes: 0.0
#             >> tests: 100.0
def lowest_score(in_dict):
# ... place implementation here

```